

The table shows the changes in internal energy of a system and the heating done to the system.

Which pair of changes requires work to be done by the system?

[1 mark]

	Change in internal energy of the system	Heating done to the system
A	Increase by 40 J	+ 20 J
B	Increase by 40 J	– 20 J
C	Decrease by 20 J	– 40 J
D	Decrease by 20 J	+ 40 J

☐☐☐☐

Do not write

A liquid with specific heat capacity c is heated in a continuous flow system. The power input to the system is P .
In a second continuous flow system, the same rate of flow is used and the same temperature rise is observed.

What are possible features of the second system?

[1 mark]

	Specific heat capacity	Power input	
A	$0.5c$	$2P$	☐
B	c	$4P$	☐
C	$2c$	$2P$	☐
D	$4c$	$0.5P$	☐

0.50 kg of water initially at 20 °C is heated by a 3.0 kW heater for 2.0 minutes without energy losses.

specific heat capacity of water = 4.2 kJ kg⁻¹ K⁻¹

specific latent heat of vaporisation of water = 2260 kJ kg⁻¹

The water reaches

[1 mark]

A a temperature below 100 °C.

☐

B 100 °C but no water is boiled away.

☐

C 100 °C and some water is boiled away.

☐

D 100 °C and all of the water is boiled away.

☐

Don't write

1	0
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A manufacturer heats milk so that its temperature rises from $6\text{ }^{\circ}\text{C}$ to $72\text{ }^{\circ}\text{C}$.
 55 Mg of milk are processed over a 24 hour period. Energy losses are negligible.

specific heat capacity of milk = $3.9\text{ kJ kg}^{-1}\text{ K}^{-1}$

What is the average power needed to process the milk?

[1 mark]

A 16.4 kW

☐

B 164 kW

☐

C 1.64 MW

☐

D 164 MW

☐

01

Balloons carry instruments into the upper atmosphere to help with weather forecasting. One balloon is inflated at ground level using 0.58 kg of helium. The inflated balloon is released and expands as it ascends. Eventually it bursts and the instruments return to the ground.

01.1

When it is inflated at ground level, the balloon experiences an upwards force of 29.3 N.

The combined mass of the skin of the balloon and the instruments is 2.40 kg.

Calculate the resultant force on the balloon as it is released at ground level.

[2 marks]

resultant force = _____ N

01.2

The speed of the balloon changes as it ascends.

Describe **two** factors that affect the speed of the balloon as it ascends.

[2 marks]

1 _____

2 _____



0 1 . 3

The temperature of the helium in the balloon changes from $25\text{ }^{\circ}\text{C}$ to $-50\text{ }^{\circ}\text{C}$ during the balloon's ascent.

Calculate the change in internal energy of the helium.

1 mol of helium has a mass of 4.0 g.

[3 marks]

change in internal energy = _____ J

0 1 . 4

The balloon expands as it ascends.

The temperature inside the balloon is always greater than the temperature outside the balloon.

Explain, using the first law of thermodynamics, how the internal energy of the helium changes during the ascent.

[3 marks]

Question 1 continues on the next page

Turn over ►



0 1 . 5

During the ascent, the volume of the balloon increases by a factor of more than a hundred.

Explain why the rate of heat transfer through the skin of the balloon increases as it ascends.

[2 marks]

12



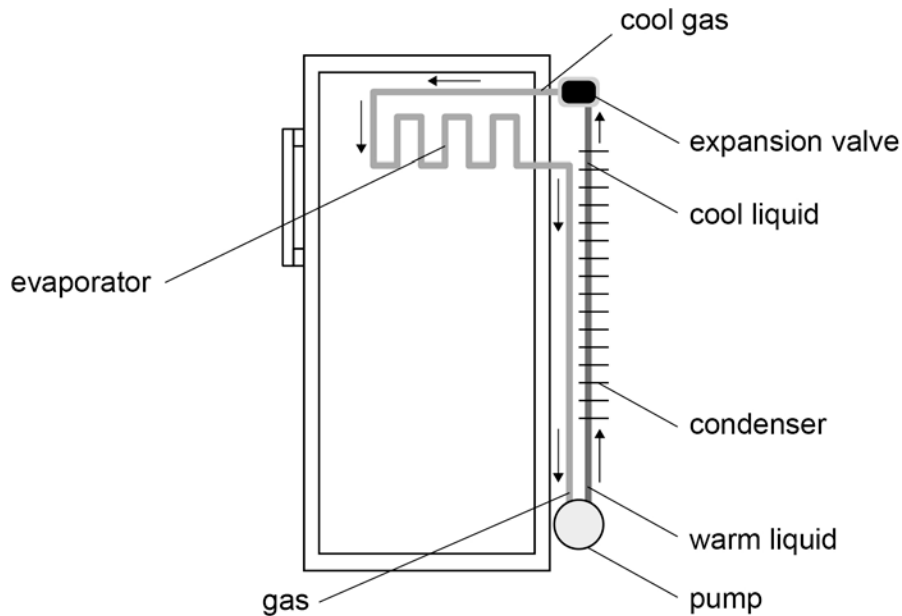
Don't write

0 2

Figure 1 shows the main parts of a refrigerator.

A fluid is pumped continuously around pipes inside and at the back of the refrigerator. The fluid transfers energy from inside the refrigerator to the surroundings.

Figure 1



- The fluid is a liquid as it reaches the expansion valve. It enters the evaporator where it changes into a cool gas and expands.
- The cool gas absorbs energy from the refrigerator contents by thermal conduction through the walls of the pipe.
- The gas turns back into a warm liquid when it is pumped into the condenser.
- This liquid cools as it travels through the condenser and energy is transferred to the surroundings.

0 2 . 1

Explain why there is an energy transfer to the fluid as it changes state in the evaporator.

[3 marks]

Question 2 continues on the next page

Turn over ►



0 2 . 2

The temperature of the refrigerator's surroundings varies throughout the day. **Figure 2** shows the variation of temperature with time for the surroundings. The normal operating temperature inside the refrigerator is a constant $5.0\text{ }^{\circ}\text{C}$.

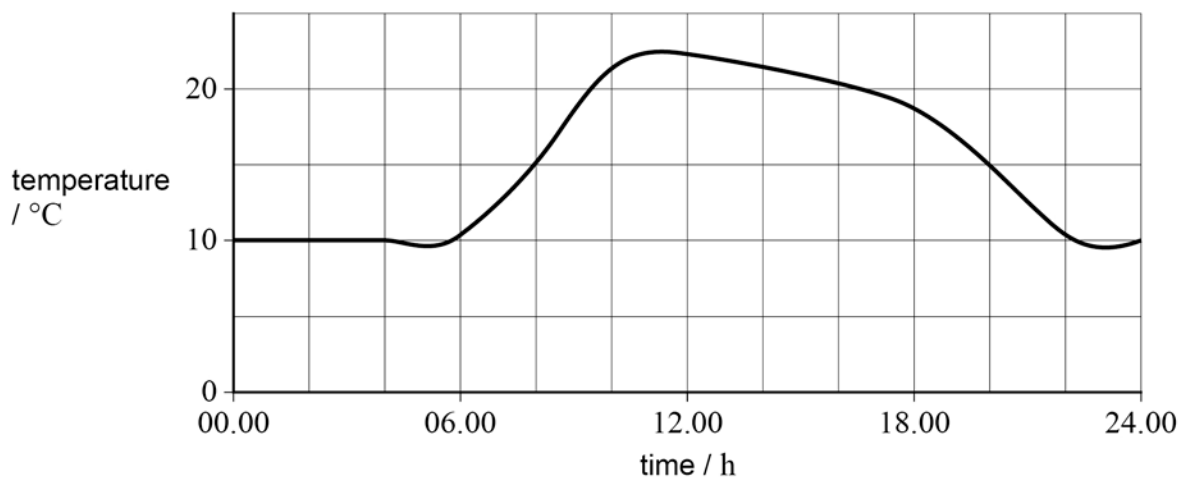
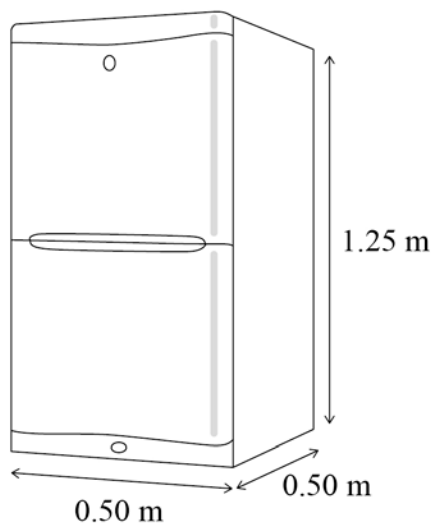
Figure 2

Figure 3 shows the dimensions of the refrigerator.

Figure 3

All faces of the refrigerator have a U-value of $0.66 \text{ W m}^{-2} \text{ K}^{-1}$.

Show that the average rate of energy conduction into the refrigerator from the surroundings is approximately 20 W.

[3 marks]

0 2 . 3

6.5 kg of food is put into the refrigerator.

The food is at an initial temperature of 21°C and has an average specific heat capacity of $3900 \text{ J kg}^{-1} \text{ K}^{-1}$.

The temperature of the food decreases to 5.0°C in 4.0 h.

Calculate, in W, the average rate of heat transfer from the food.

[2 marks]

average rate of heat transfer = _____ W

Question 2 continues on the next page

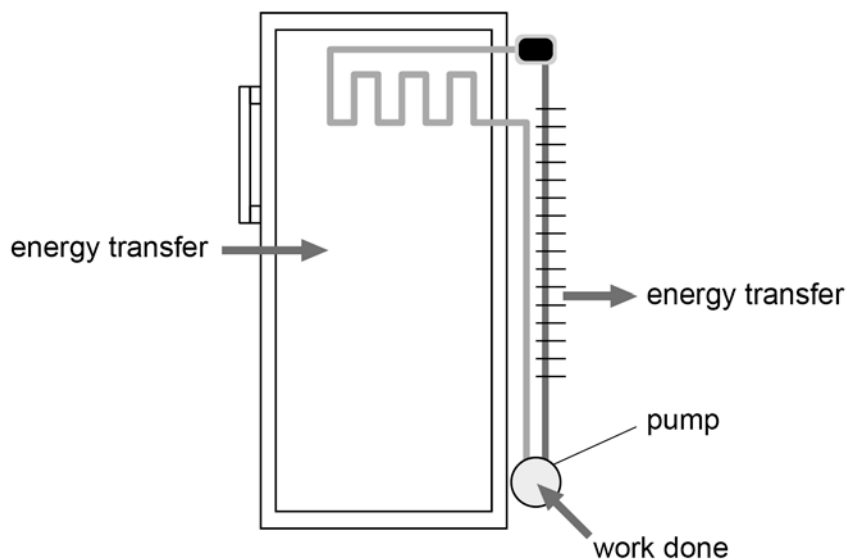


0 2 . 4

Figure 4 shows the refrigerator. The total internal energy of the refrigerator is affected in three ways:

- work is done by the pump on the fluid in the pipes at the rate of 60 W
- energy is transferred **into** the refrigerator through its walls
- energy is transferred **from** the refrigerator to the surroundings by the condenser.

Figure 4



Calculate the total rate of energy transfer from the condenser to the surroundings as the refrigerator is cooling the food to 5.0 °C.

[2 marks]

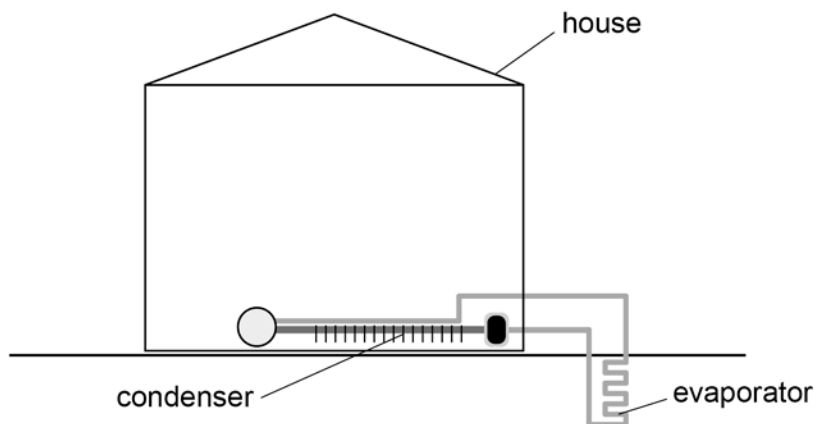
rate of energy transfer = _____ W



0 2 . 5

An engineer suggests heating a house using a reversed refrigerator system. He suggests that the evaporator is in the ground outside the house and the condenser is inside the house as shown in **Figure 5**.

Figure 5



The engineer thinks that using this system will be more energy efficient than using normal electrical heating for the house.

Discuss whether the engineer is correct.

[2 marks]

Turn over for the next question



Do not write

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0**6**

Which process will increase the kinetic energies of the particles involved?

[1 mark]

A boiling a liquid at its boiling point

☐

B compressing a gas without heat transfer

☐

C expanding a gas without heat transfer

☐

D melting a solid at its melting point

☐

Energy is transferred at the rate of 6.0 kW to a water sample of mass 4.0 kg. The water does not boil. The specific heat capacity of water is $4.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

What is the initial rate of temperature rise of the water?

[1 mark]

A $3.6 \times 10^{-4} \text{ K s}^{-1}$

☐

B $1.3 \times 10^3 \text{ K h}^{-1}$

☐

C 1.3 K h^{-1}

☐

D $3.6 \times 10^3 \text{ K s}^{-1}$

☐

Turn over ►



Questions **08** and **09** are about the equation $\Delta U = Q + W$

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0

8

What are Q and W ?

[1 mark]

	Q	W	
A	energy input to the system by heating	work done by the system	☐
B	energy input to the system by heating	work done on the system	☐
C	energy output from the system by heating	work done by the system	☐
D	energy output from the system by heating	work done on the system	☐

What is a unit for $\frac{\Delta Q}{t}$ where t is time?

[1 mark]

A K

☐

B K s^{-1}

☐

C W

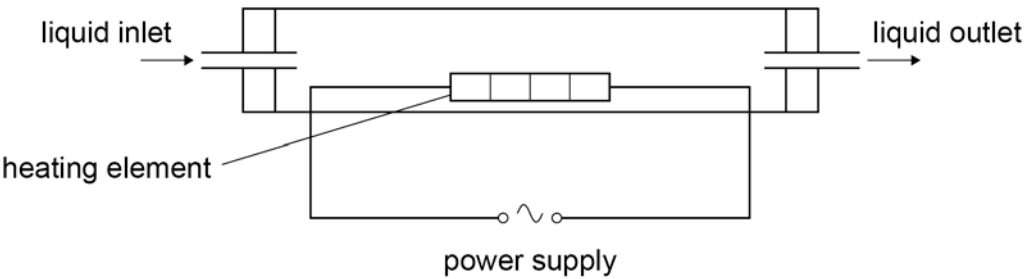
☐

D W s^{-1}

☐

Don't write

The diagram shows a liquid passing through a heater at a steady mass flow rate m . The rate of energy supply is P_1 and the rate of energy loss is P_2 . The temperature rise of the liquid is ΔT .



Which set of conditions will result in a temperature rise of $2\Delta T$?

[1 mark]

	Rate of energy supply	Rate of energy loss	Mass flow rate	
A	$2P_1$	$2P_2$	m	☐
B	$2P_1$	$0.5P_2$	m	☐
C	P_1	$0.5P_2$	$0.5m$	☐
D	P_1	$2P_2$	$0.5m$	☐

Turn over for the next question



Do not write Do not write

1 1

An experiment to measure the specific heat capacity of a material is performed. Energy losses are negligible.

Which measurements for the material and for the heater are sufficient to calculate the specific heat capacity?

[1 mark]

	Measurements for the material	Measurements for the heater	
A	Initial and final temperature Mass	Power supplied	☐
B	Initial and final temperature Volume	Energy supplied	☐
C	Initial and final temperature Mass	Voltage Current	☐
D	Initial and final temperature Volume Density	Energy supplied	☐

1**2**

Which is the fundamental (base) unit for specific heat capacity?

[1 mark]

A $\text{m}^2 \text{s}^{-4} \text{K}^{-1}$

☐

B $\text{kg m}^2 \text{s}^{-2} \text{K}^{-1}$

☐

C $\text{kg}^{-1} \text{m}^2 \text{s}^{-2} \text{K}^{-1}$

☐

D $\text{m}^2 \text{s}^{-2} \text{K}^{-1}$

☐

Don't write

1 **4** A wall **P** has thickness t and area A . The rate of heat transfer through **P** is W .

A second wall **Q** has thickness $2t$, area $\frac{A}{2}$ and twice the U-value of **P**.

The temperature difference across each wall is the same.

The rate of heat transfer through **Q** is

[1 mark]

A $\frac{W}{2}$



B W



C $2W$



D $4W$



0 8

The table shows information about four materials.

Material	Area / m²	Thickness / mm	<i>U</i>-value / W m⁻² K⁻¹
A	1.0	40	3.0
B	2.0	10	1.0
C	3.0	20	2.0
D	3.0	30	4.0

The rate of energy transfer through each material is the same.

Which material will have the greatest temperature difference across its surfaces?

[1 mark]

A ☐

B ☐

C ☐

D ☐

1

3

An insulated beaker of negligible heat capacity contains a liquid at a temperature of 10°C .
The mass of the liquid is 0.30 kg .
A metal block of mass 0.10 kg is at a temperature of 90°C .
The metal is placed into the liquid.
The metal and liquid reach thermal equilibrium at a temperature of 28°C .
The specific heat capacity of the liquid is $1700\text{ J kg}^{-1}\text{ K}^{-1}$.

What is the specific heat capacity of the metal?

[1 mark]

A $160\text{ J kg}^{-1}\text{ K}^{-1}$

☐

B $1300\text{ J kg}^{-1}\text{ K}^{-1}$

☐

C $1500\text{ J kg}^{-1}\text{ K}^{-1}$

☐

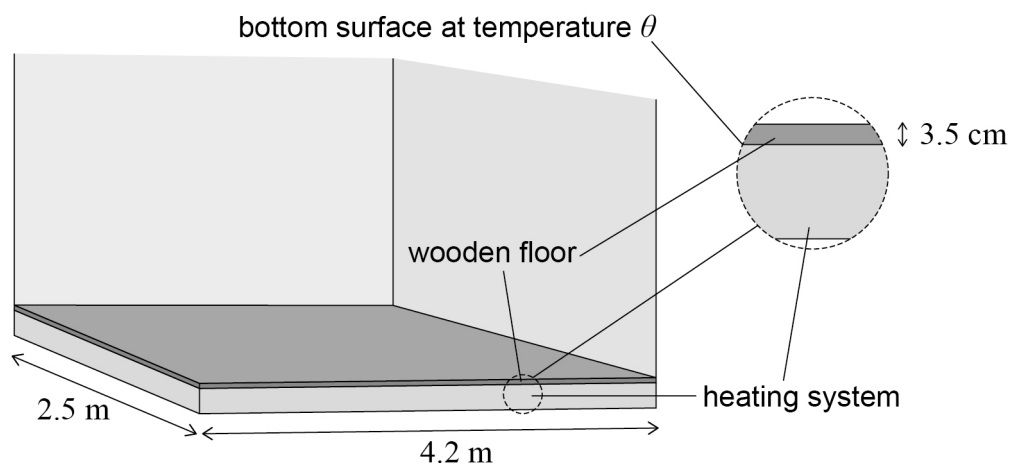
D $1800\text{ J kg}^{-1}\text{ K}^{-1}$

☐

03

A room is heated by a heating system under the floor.
The room has a wooden floor, as shown in **Figure 4**.
The floor is 2.5 m wide, 4.2 m long and 3.5 cm thick.

Figure 4



The heating system keeps all points on the bottom surface of the floor at temperature θ .

θ can be changed by adjusting the power output of the heating system.

03.1

Initially, θ is $45\text{ }^{\circ}\text{C}$ and the temperature at the upper surface of the floor is $22\text{ }^{\circ}\text{C}$.
Energy is transferred by conduction through the floor at a rate of 1.2 kW.

Calculate the thermal conductivity of the wood used to make the floor.
State an appropriate SI unit for your answer.

[4 marks]

thermal conductivity = _____ unit = _____



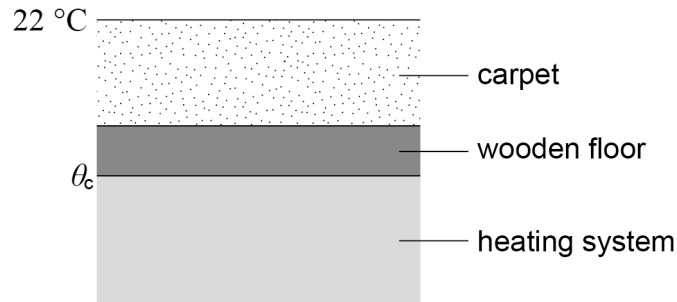
0 3 . 2

The whole of the floor is now covered with a carpet. The combined U-value of the floor and carpet is $2.1 \text{ W m}^{-2} \text{ K}^{-1}$.

The power output of the heating system is adjusted until the rate of energy transfer through the floor and carpet is 1.2 kW.

The temperature of the top of the carpet is 22°C as shown in the cross-sectional view of the floor in **Figure 5**.

Figure 5



Calculate θ_c , the new temperature of the bottom of the wooden floor.

[2 marks]

$\theta_c =$ _____ $^\circ\text{C}$

0 3 . 3

Some of the energy is wasted heating the ground underneath the heating system.

Suggest how the use of a carpet on the floor affects the wasted heat transfer from the heating system.

[2 marks]

04.1

Estimate the theoretical maximum value of the radius of a silicon nucleus using the closest approach of a 6.5 MeV alpha particle.

proton number of silicon = 14

[4 marks]

radius of silicon nucleus = _____ m

04.2

Explain why the value obtained in Question 04.1 is only an estimate of the theoretical maximum value of the radius.

[1 mark]

Table 1 contains data about two nuclides **X** and **Y**.

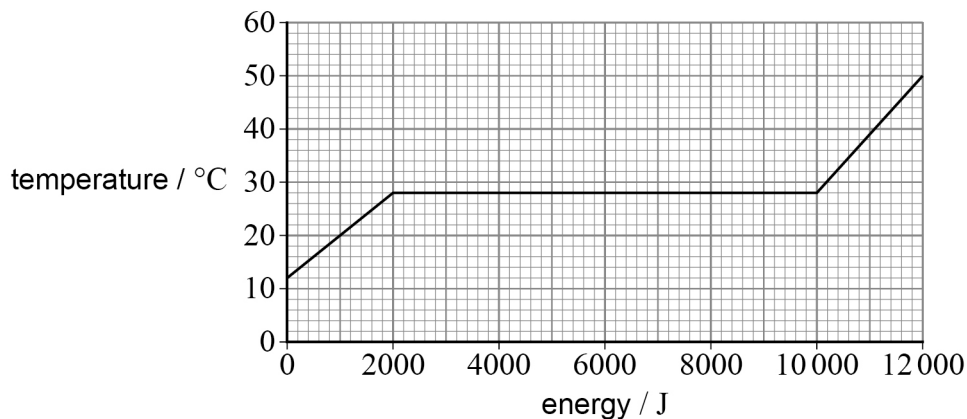
Table 1

Nuclide	Number of protons in nucleus	Number of neutrons in nucleus
X	16	24
Y	20	20



Don't write

The graph shows how the temperature of 500 g of a metal changes as energy is transferred to the metal.



What is the specific latent heat of the metal?

[1 mark]

A 0.16 kJ kg^{-1}

☐

B 12 kJ kg^{-1}

☐

C 16 kJ kg^{-1}

☐

D 24 kJ kg^{-1}

☐

Don't write

The first law of thermodynamics for a system can be written as:

$$\Delta U = Q + W$$

The energy input to a system by heating is 800 J, and the system does 600 J of work.

What is ΔU ?

[1 mark]

A 1400 J

☐

B 200 J

☐

C -200 J

☐

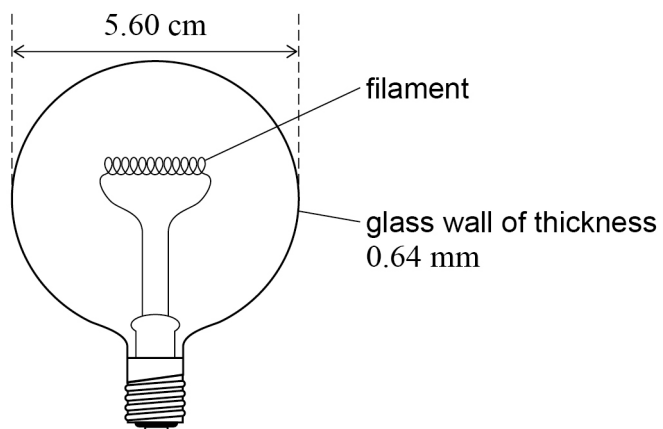
D -1400 J

☐

0 2

Figure 2 shows a filament lamp with a total power output of 60.0 W .
The glass wall of the lamp has a uniform thickness of 0.64 mm .

Figure 2



Assume the lamp to be a glass sphere of external diameter 5.60 cm .

The lamp is turned on and reaches thermal equilibrium.

The temperature difference across the glass wall is now 1.2 K .

0 2

1

Calculate the percentage of the power output of the lamp that transfers through the glass by conduction.

thermal conductivity of glass = $1.10 \text{ W m}^{-1} \text{ K}^{-1}$

[3 marks]

percentage of power output = _____



0 2 . 2

At equilibrium, the mean temperature of the glass wall is 131 °C.

Calculate the energy needed to increase the temperature of the glass from 20 °C to 131 °C.

density of glass = $2.5 \times 10^3 \text{ kg m}^{-3}$
specific heat capacity of glass = $0.84 \text{ kJ kg}^{-1} \text{ K}^{-1}$

[3 marks]

energy = _____ J

6

Turn over for the next question

Turn over ►



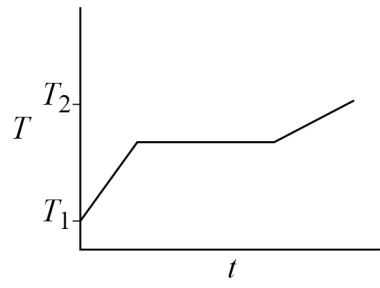
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0 8

A material is initially solid at a temperature of T_1 .

The material is heated with a constant power of P . It melts and the temperature of the liquid increases to T_2 .

The graph shows the variation of the temperature T with time t for the material.



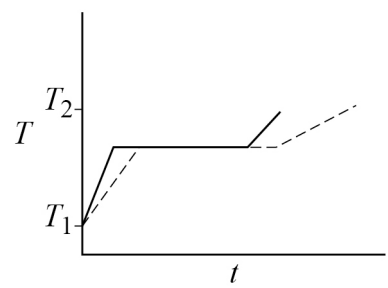
The same mass of the same material is now heated with a constant power of $2P$.

Which graph shows the new variation of T with t ?

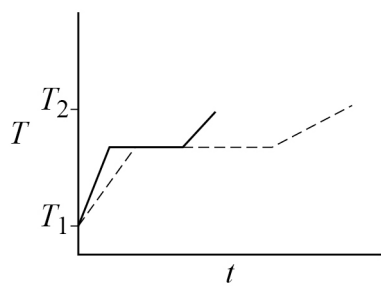
The dashed lines show the shape of the original graph.

[1 mark]

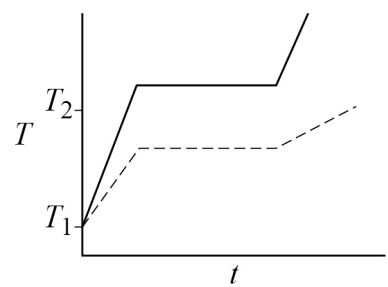
A



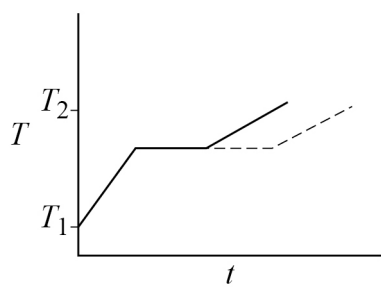
B



C



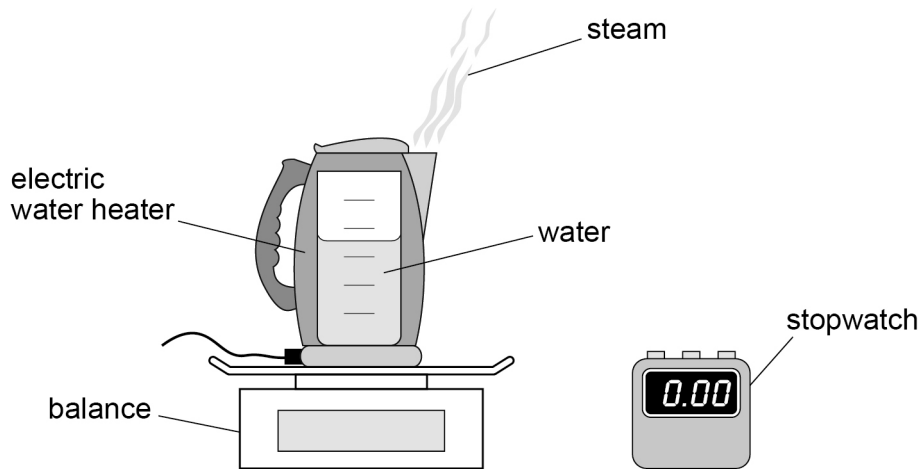
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- A ☐
- B ☐
- C ☐
- D ☐

Don't write

A student does an experiment to estimate the specific latent heat L of water.



An electric water heater is placed on a balance and switched on.

The water heater uses a constant power of P .

The student waits until the water is boiling. He then starts the stopwatch and measures the time t taken for the mass of water to decrease by m .

The student estimates L using:

$$mL = Pt$$

Which error would result in an overestimate of L ?

[1 mark]

A underestimating t

☐

B overestimating m

☐

C ignoring energy losses from the water heater

☐

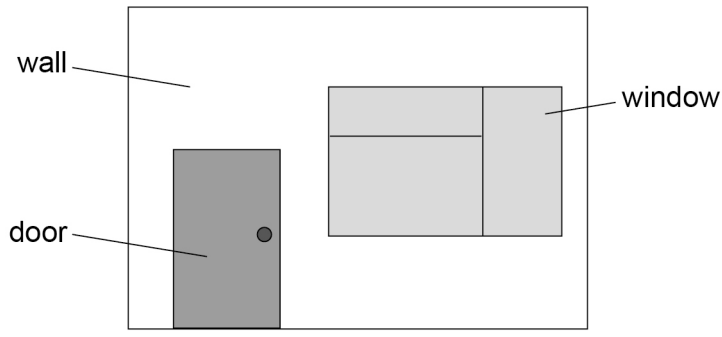
D ignoring the mass of drops of water emitted with the steam

☐


Do not write

1 0

The front of a house has three parts: a wall, a window and a door.



The table shows the U-values and the areas for the three parts.

Part	U-value / $\text{W m}^{-2} \text{K}^{-1}$	Area / m^2
wall	1.0	8.0
window	1.5	2.3
door	2.2	1.6

Which row shows the parts in order from the smallest rate of heat transfer to the largest rate of heat transfer?

[1 mark]

- A** wall, window, door ☐
- B** wall, door, window ☐
- C** window, door, wall ☐
- D** window, wall, door ☐

Turn over for the next question

Turn over ►



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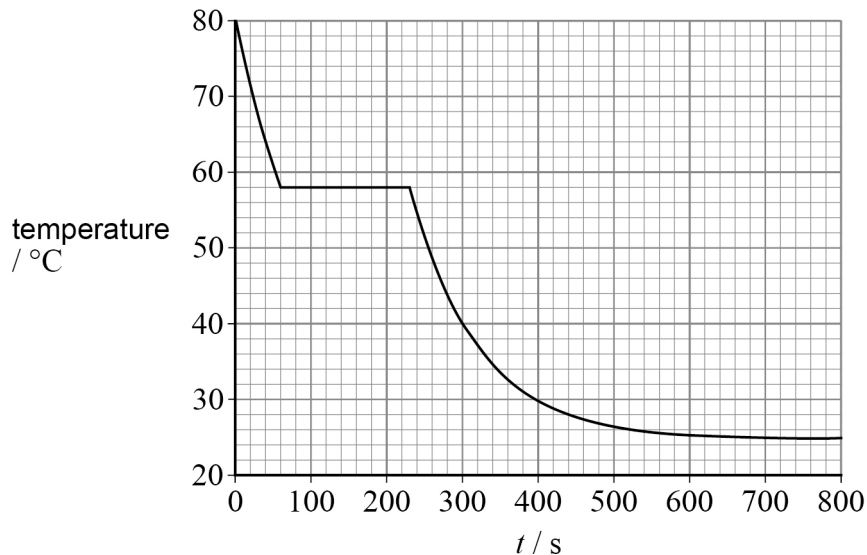
0 1

A test tube contains a sample **S** of a liquid at $80.0\text{ }^{\circ}\text{C}$.

A student records the temperature of **S** using a temperature sensor and data logger. At time $t = 0$, **S** starts to cool down. It turns into a solid before cooling to room temperature.

Figure 1 shows the variation of the temperature of **S** with t .

Figure 1

**0 1 . 1**

Suggest why the student uses a temperature sensor and data logger rather than a thermometer and stopclock when recording the temperature in this experiment.

[1 mark]

0 1 . 2

Explain, in terms of energy transfer, why the temperature of **S** does not change between $t = 60\text{ s}$ and $t = 230\text{ s}$.

[2 marks]



Table 1 shows some of the properties of **S**.

Table 1

mass / kg	6.60×10^{-2}
specific heat capacity of liquid / $\text{J kg}^{-1} \text{K}^{-1}$	2.80×10^3
specific latent heat of fusion / J kg^{-1}	2.64×10^5
specific heat capacity of solid / $\text{J kg}^{-1} \text{K}^{-1}$	2.10×10^3

0 1 . 3 Calculate the total energy transferred from **S** during the time shown in **Figure 1**.

[4 marks]

energy transferred = _____ J

0 1 . 4 Determine, using **Figure 1**, the rate of temperature decrease of **S** when $t = 300 \text{ s}$.

[2 marks]

rate = _____ K s^{-1}

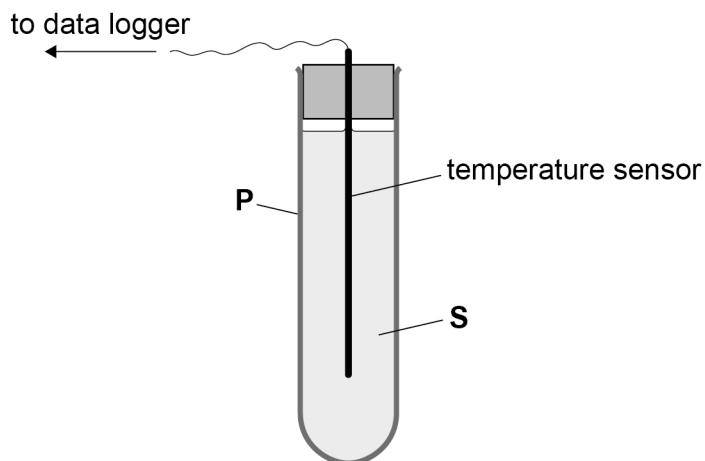
Question 1 continues on the next page

Turn over ►



0 1 . 5 Figure 2 shows **S** in the test tube.

Figure 2



P is a point on the **outside** of the test tube as shown in **Figure 2**.

The test tube is made from glass of thickness 1.2 mm.

The thermal conductivity of the glass is $1.14 \text{ W m}^{-1} \text{ K}^{-1}$.

Energy is transferred by conduction from **S** through an area of $8.0 \times 10^{-3} \text{ m}^2$ of the glass. Assume that all the energy transferred from **S** is by conduction through this area.

When $t = 400 \text{ s}$ the rate of temperature decrease of **S** is 0.055 K s^{-1} .

Determine the temperature of **P** when $t = 400 \text{ s}$.

[3 marks]

temperature of **P** = _____ °C

Do not write

1**2**

An isolated system initially has an internal energy U_0 .

An amount of energy Q is removed from the system by cooling.
The internal energy increases to U_1 .

What is the work done on the system?

[1 mark]

A $U_0 + U_1 - Q$

☐

B $U_1 + Q - U_0$

☐

C $U_0 + Q - U_1$

☐

D $U_0 - U_1 - Q$

☐

Turn over for the next question

Turn over ►



Don't write

1**4**

What are the fundamental (base) units of U-value?

[1 mark]

A $\text{kg m}^2 \text{s}^{-3} \text{K}^{-1}$

☐

B $\text{kg s}^{-2} \text{K}^{-1}$

☐

C $\text{kg s}^{-3} \text{K}^{-1}$

☐

D $\text{kg m}^{-1} \text{s}^{-3} \text{K}^{-1}$

☐

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0 2

The boiling point of water is $100\text{ }^{\circ}\text{C}$.

Water is heated to its boiling point and then changes state to become water vapour.

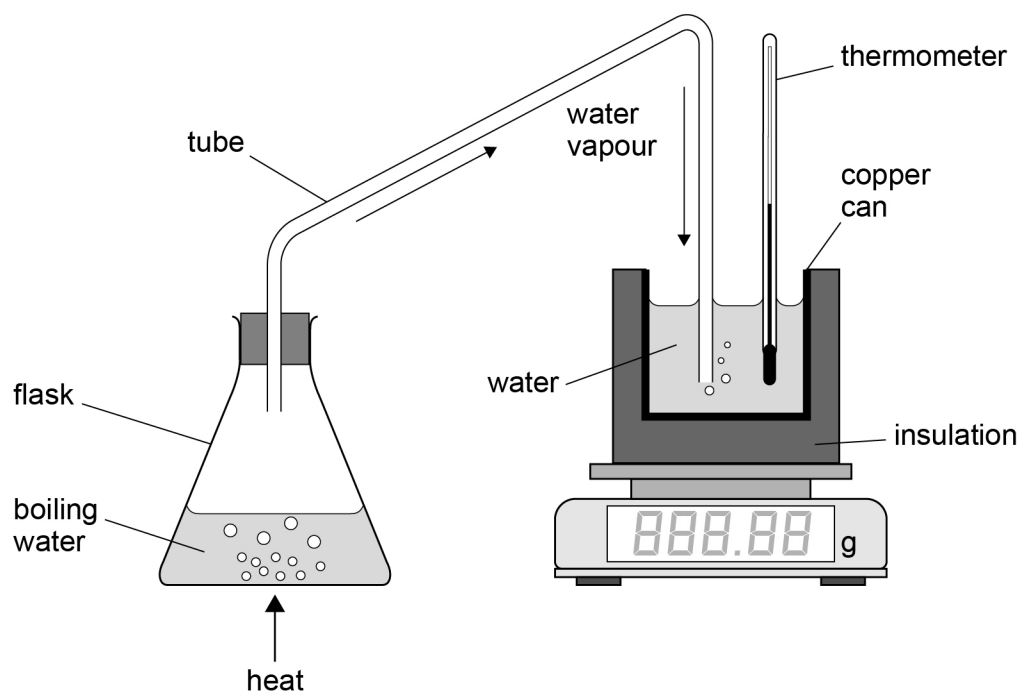
0 2 . 1

Describe what happens to the kinetic energies and potential energies of the water particles during the change of state.

[2 marks]

Figure 2 shows the arrangement that a student uses to determine the specific latent heat of water.

Figure 2



Throughout the experiment the temperature of the copper can is equal to the temperature of the water in the can.

Just as the water in the flask begins to boil, the student measures the initial temperature of the water in the can and the initial mass of the water in the can.

There is no liquid water in the water vapour. Water vapour at $100\text{ }^{\circ}\text{C}$ enters the water in the can. This causes the temperature of the water and can to increase. The liquid water formed from the water vapour and the water in the can reach the same temperature.

Question 2 continues on the next page

Turn over ►



The experiment is stopped when the water and the can reach a temperature of 45°C .

The water vapour transfers a total of 36 kJ of energy to the original water in the can and to the can.

The specific heat capacity of water is $4200\text{ J kg}^{-1}\text{ K}^{-1}$.

0 2 . 2

Determine the initial temperature of the water and can.

initial mass of water in can = 0.250 kg

mass of can = 0.350 kg

specific heat capacity of copper = $385\text{ J kg}^{-1}\text{ K}^{-1}$

[3 marks]

initial temperature = _____ $^{\circ}\text{C}$

0 2 . 3

During the experiment the mass of water in the can increases by 0.015 kg .
This increase in mass is due to the water vapour changing state.

Determine the specific latent heat of this change of state.

[2 marks]

specific latent heat = _____ J kg^{-1}



0 2 . 4

State and explain **two** changes to this experiment that would improve the accuracy of the value determined for the specific latent heat.

[2 marks]

1 _____

2 _____

9

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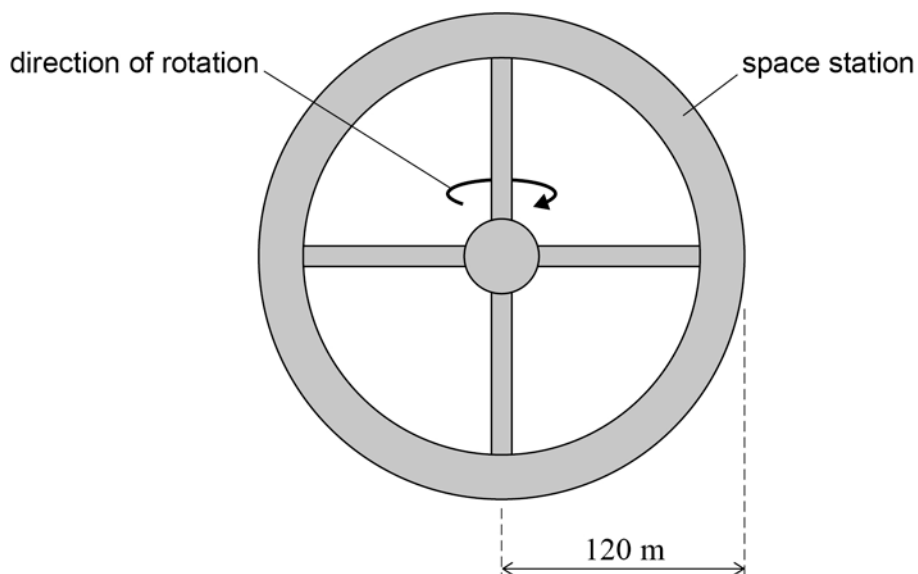
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0	2
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Figure 1 shows an idea for a future space station. The station rotates about its axis to simulate gravity in its outer section.



0	2
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1

The space station has a moment of inertia of $5.2 \times 10^{10} \text{ kg m}^2$.
It is accelerated from rest with a torque of $4.4 \times 10^4 \text{ N m}$ for 3.0 days.

Show that angular speed of the space station is approximately 0.22 rad s^{-1} .

[3 marks]

0	2
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2

Calculate the work done accelerating the space station from rest to an angular velocity of 0.22 rad s^{-1} .

[1 mark]

work done = _____ J

0 2 . 3

An astronaut moves from the centre of the space station to the outer rim.
Discuss the effect that the change in position of the astronaut will have on
the angular velocity of the space station.

[3 marks]

03 . 1

A conventional pump storage system has a power output of 8.0×10^7 W.
The height difference between its reservoir and its turbine is 610 m.
The generation process is 95% efficient.

Calculate the mass of water passing through the turbine each second.

[4 marks]

mass per second = _____ kg

03 . 2

The annual output of electrical energy is 180 GW h.

Show that the average time for which the power station operates is approximately 6 hours per day.

[3 marks]

03 . 3

Identify **one** advantage of using a pump storage system to generate electricity.

[1 mark]

0 4

The volume of air in a fully expanded human lung is $2.5 \times 10^{-3} \text{ m}^3$.
The pressure of the air in the lung is $1.0 \times 10^5 \text{ Pa}$ and its temperature is 37°C .

0 4 . 1

Determine the number of moles of air in the lung.

[2 marks]

number of moles _____.

0 4 . 2

Calculate the average kinetic energy of an air molecule in the lung.

[2 marks]

average kinetic energy _____ J

0 4 . 3

Air is a mixture of oxygen and nitrogen molecules. The mass of an oxygen molecule is greater than the mass of a nitrogen molecule.

Explain the effect this has on the mean square speeds of the oxygen and nitrogen molecules in the lungs.

[3 marks]

0 5

The mass of a nucleus ${}^4_2\text{He}$ is 4.00151 u. The mass of a proton is 1.00728 u and the mass of a neutron is 1.00867 u.
 $1 \text{ u} = 1.661 \times 10^{-27} \text{ kg}$

0 5

. 1

Calculate the mass defect Δm of this nucleus in a.m.u.

[3 marks]

$\Delta m =$ _____.

0 5

. 2

Calculate the binding energy per nucleon of this nucleus.

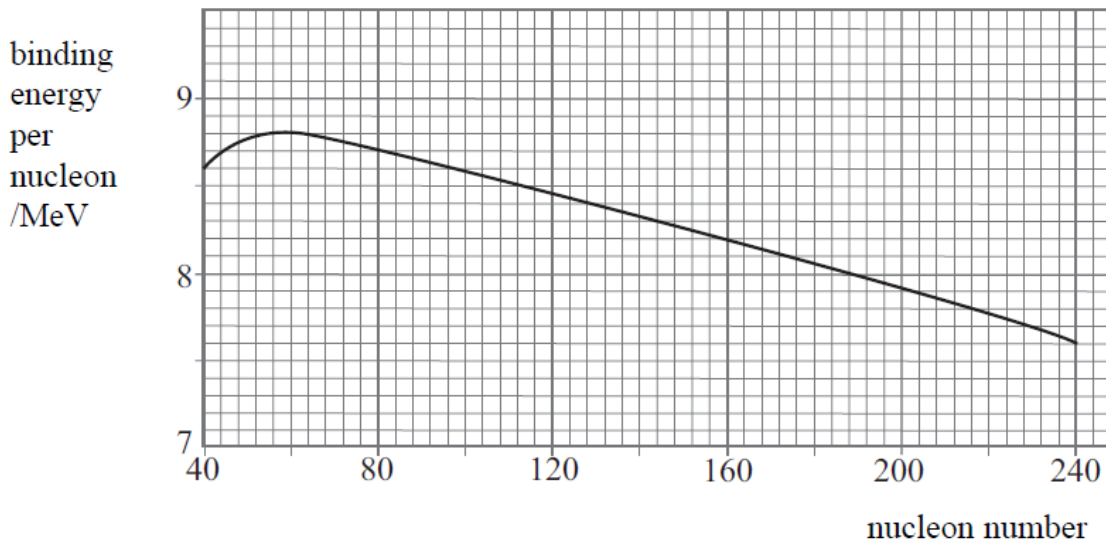
[2 marks]

Binding energy per nucleon = _____.

Turn over for the next question

Figure 1 shows an enlarged portion of a graph indicating how the binding energy per nucleon of various nuclides varies with their nucleon number.

Figure 1



0 5 . 3

A textbook states that "When fission of Uranium 235 takes place, the nucleus splits into two roughly equal parts and approximately 200 MeV of energy is released".

Comment on the validity of this energy value, using **Figure 1** to help you.
Explain your reasoning.

[3 marks]

0 6

A student immerses a 2.0 kW electric heater in a beaker of water. The heater is switched on for 120 s. At this point the heater is switched off and immediately removed from the water.

The data below was collected during the experiment.

initial mass of beaker 25 g

initial mass of beaker and water 750 g

initial temperature of water 20 °C

final temperature of water 95 °C

0 6

1

Calculate the specific heat capacity of water.
Give an appropriate unit.

[4 marks]

Specific heat capacity _____ Unit _____.

0 6

2

State and explain **two** ways in which the student could improve the accuracy of the value obtained for the specific heat capacity.

[4 marks]

07

The first law of thermodynamics can be represented by $\Delta U = Q + W$.

07

. 1

State and explain, with reference to the equation, **two** ways in which the internal energy of a gas can be decreased.

[4 marks]

A volume of 20 m^3 of exhaust gas from a diesel engine leaves the exhaust pipe at a pressure of $1.0 \times 10^5 \text{ Pa}$. The gas is cooled by the surrounding atmosphere, which is also at a pressure of $1.0 \times 10^5 \text{ Pa}$, and, as a result, the exhaust gas contracts to half its volume.

07

. 2

Calculate the work done by the atmosphere on the gas during this contraction.

[1 mark]

07 . 3

During the cooling 4.9 MJ of energy is transferred to the atmosphere.

Calculate the change in internal energy of the gas using the first law of thermodynamics.

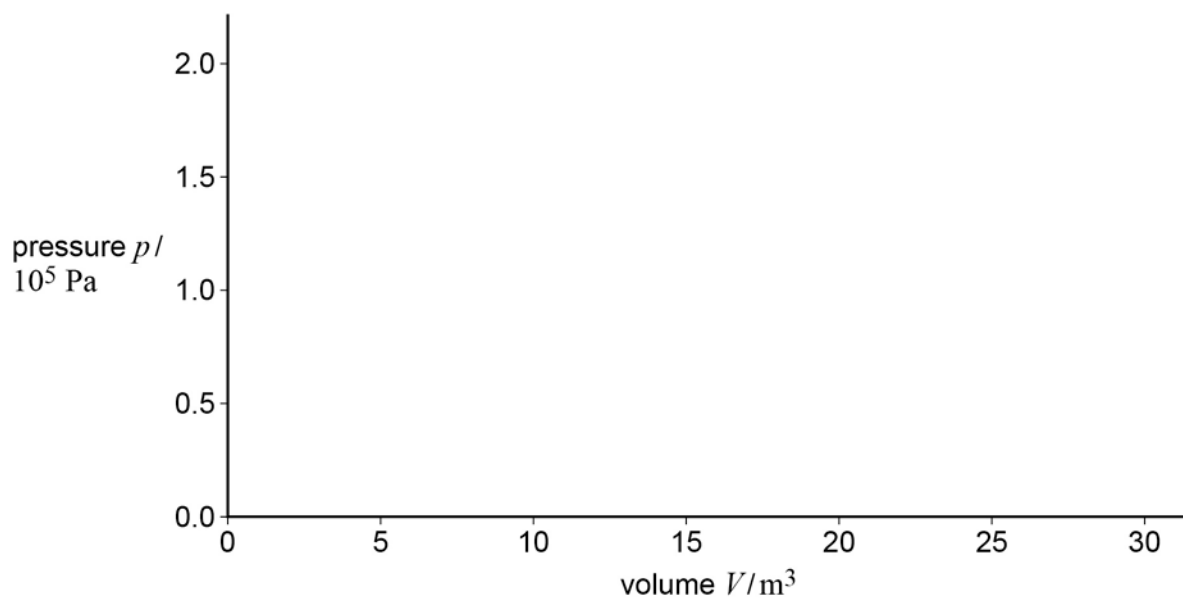
[1 mark]

07 . 4

Use **Figure 2** to represent this process as a p - V diagram, showing the direction of the process.

[2 marks]

Figure 2



Turn over for the next question

08

This question is about the kinetic theory of gases.

08

. 1

State **two** assumptions of the kinetic theory of gases.

[2 marks]

Assumption 1

Assumption 2

08

. 2

Show that the mean kinetic energy of one molecule of an ideal gas at a temperature of 21 °C is about 6×10^{-21} J.

Boltzmann constant = 1.4×10^{-23} J K⁻¹

[2 marks]

Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C**, and **D**. For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

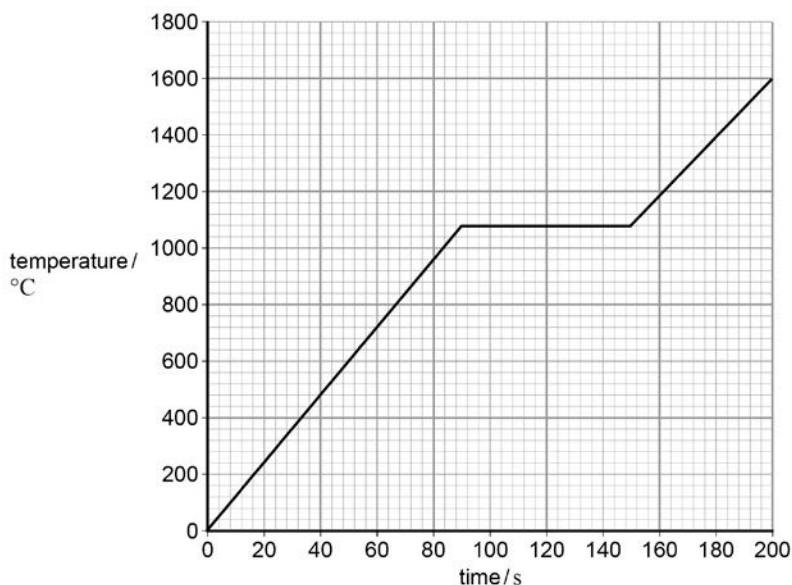


If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



0 9

A 60 g solid metal sample is heated by an electric heater. The heater operates at a voltage of 230 V and with a current of 1.0 A. The temperature of the sample changes as shown below:



What is the specific latent heat of vaporisation of the metal?

[1 mark]

A $\frac{1}{230} \text{ J kg}^{-1}$

☐

B $\frac{1}{23} \text{ J kg}^{-1}$

☐

C 230 J kg⁻¹

☐

D 230 × 10³ J kg⁻¹

☐

1 0

A gas molecule of mass m moving at velocity u collides at right angles with the side of a container and rebounds elastically.

Which of the following statements concerning the motion of the molecule is **incorrect**?

[1 mark]

- A** The magnitude of the change in momentum of the molecule is zero. ☐
- B** The magnitude of the change in momentum of the molecule is $2mu$. ☐
- C** The force exerted by the molecule on the side of the container is equal to the force exerted by the container on the molecule. ☐
- D** The change in kinetic energy of the molecule is zero. ☐

1 1

The mean kinetic energy of the molecules in a gas is:

[1 mark]

- A** directly proportional to the absolute temperature ☐
- B** directly proportional to the square root of temperature ☐
- C** independent of absolute temperature ☐
- D** inversely proportional to the absolute temperature ☐

1 2

A liquid flows continuously through a chamber that contains an electric heater. When the steady state is reached, the liquid leaving the chamber is at a higher temperature than the liquid entering the chamber. The difference in temperature is ΔT

Which of the following will increase ΔT with no other change?

[1 mark]

- A** Increasing the volume flow rate of the liquid ☐
- B** Changing the liquid to one with a lower specific heat capacity ☐
- C** Using a heating element with a higher resistance ☐
- D** Changing the liquid to one that has a higher density ☐

1 3

The temperature of a hot liquid in a container falls at a rate of 2 K per minute just before it begins to solidify. The temperature then remains steady for 20 minutes by which time all the liquid has all solidified. Assume the rate of heat loss remains constant throughout this process.

What is the quantity $\frac{\text{Specific heat capacity of the liquid}}{\text{Specific latent heat of fusion}}$?

[1 mark]

A $\frac{1}{40} \text{ K}^{-1}$ ☐

B $\frac{1}{10} \text{ K}^{-1}$ ☐

C 10 K^{-1} ☐

D 40 K^{-1} ☐

1 4

A fixed mass of gas occupies a volume V . The temperature of the gas increases so that the root mean square velocity of the gas molecules is doubled.

What will the new volume be if the pressure remains constant?

[1 mark]

A $\frac{V}{2}$ ☐

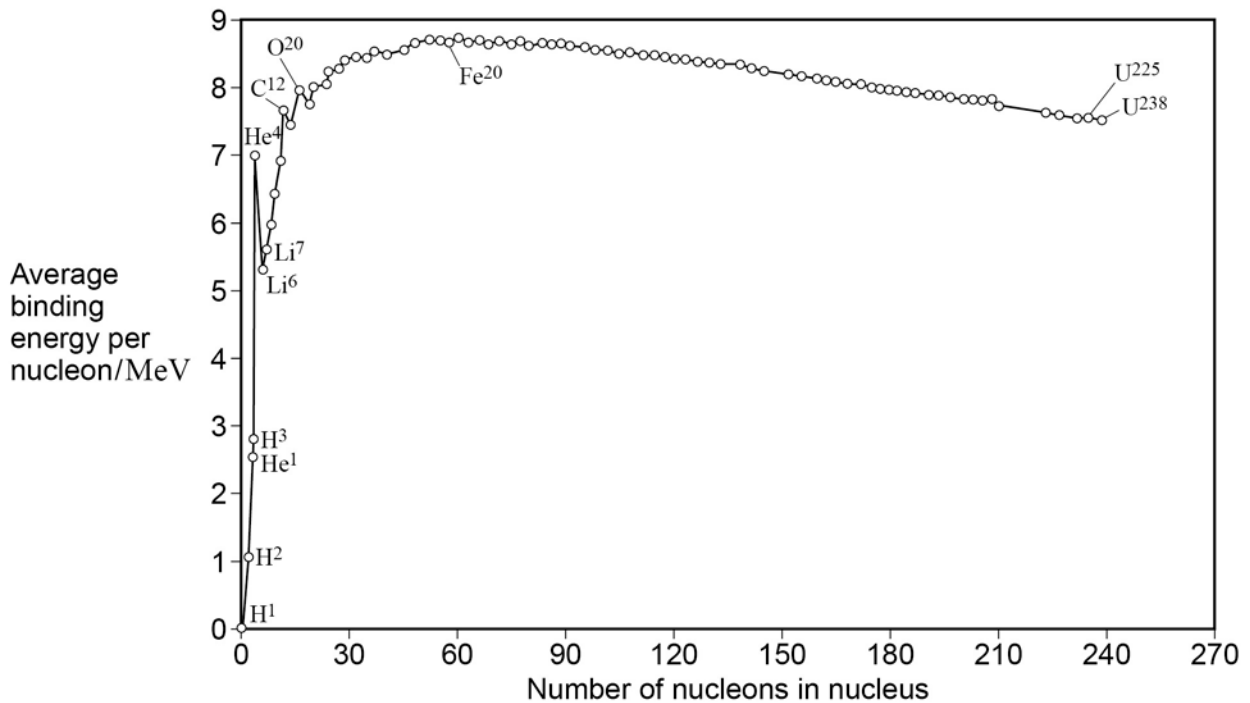
B $\frac{V}{\sqrt{2}}$ ☐

C $2V$ ☐

D $4V$ ☐

1 5

The graph shows how the binding energy per nucleon varies with the nucleon number for stable nuclei.



What is the approximate total binding energy for a nucleus of ${}^{184}_{74}\text{W}$?

[1 mark]

- A 1.28 pJ ☐
- B 94.7 pJ ☐
- C 103 pJ ☐
- 230 pJ ☐

1 6

Which of the following is the ratio of the nuclear radii of elements with mass numbers 216 and 125?

[1 mark]

- A 6 : 5 ☐
- B 216:125 ☐
- C $\sqrt{216} : \sqrt{125}$ ☐
- D $216^3 : 125^3$ ☐

1	7
---	---

A perfect gas at $27\text{ }^{\circ}\text{C}$ is heated at constant pressure so as to double its volume.

What is the increase in temperature of the gas?

[1 mark]

A $54\text{ }^{\circ}\text{C}$ ☐

B $300\text{ }^{\circ}\text{C}$ ☐

C $327\text{ }^{\circ}\text{C}$ ☐

D $600\text{ }^{\circ}\text{C}$ ☐

1	8
---	---

n moles of an ideal gas is at pressure P .

What is the pressure of $4n$ moles of the same gas at three times the temperature and half the volume?

[1 mark]

A $P/6$ ☐

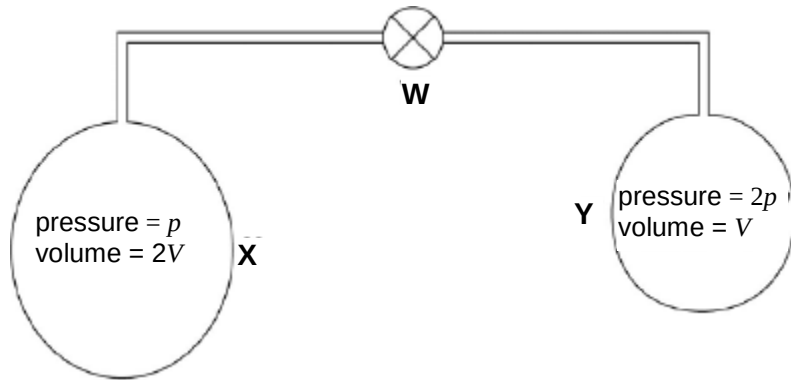
B $6P$ ☐

C $12P$ ☐

D $24P$ ☐

1 9

X and **Y** are two gas bottles that are connected by a tube that has negligible volume compared with the volume of each bottle.



Initially the valve **W** is closed.

X has a volume $2V$ and contains hydrogen at a pressure of p .

Y has a volume V and contains hydrogen at a pressure of $2p$.

X and **Y** are both initially at the same temperature.

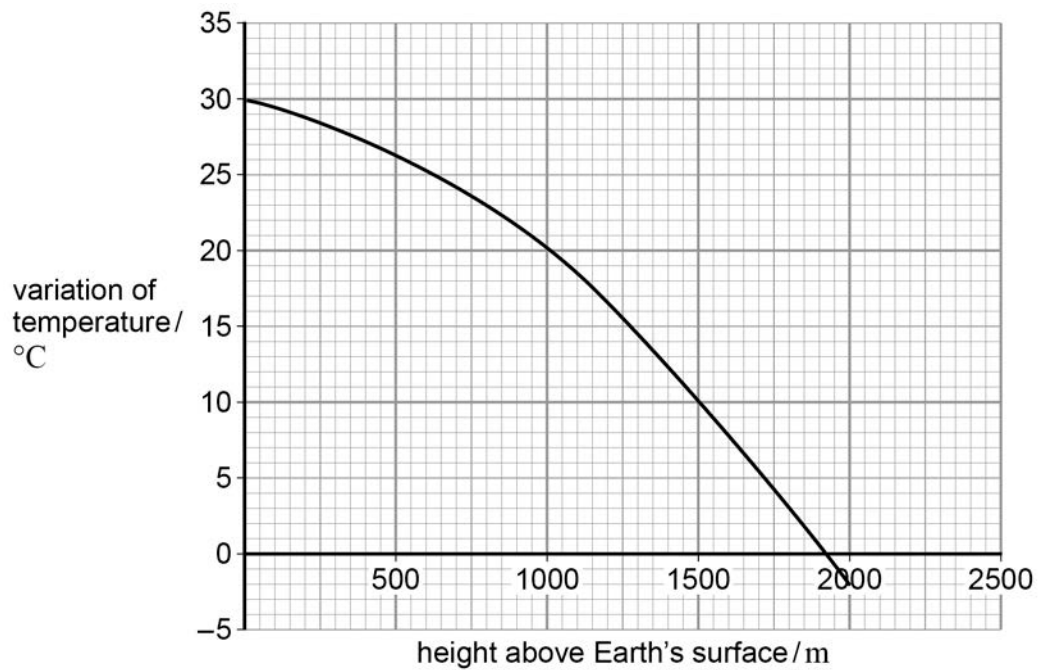
W is now opened. Assuming that there is no change in temperature.
What is the new gas pressure?

[1 mark]

A	$\frac{2}{3}p$	☐
B	$\frac{5}{3}p$	☐
C	$\frac{4}{3}p$	☐
D	$\frac{3}{2}p$	☐

2 0

A spherical balloon, of radius 12 cm, is filled with helium at a pressure of 14 kPa. The balloon is released from the surface of the earth and travels to a height of 1500 m. The graph below shows the variation of temperature with height above the Earth's surface.



Calculate the pressure in the balloon at 1500 m.

[1 mark]

- A** 4.2 Pa ☐
- B** 4.6 Pa ☐
- C** 4500 Pa ☐
- D** 4700 Pa ☐

2	1
---	---

What is the mass defect of the ${}^7_3\text{Li}$ nucleus?

Use the following data:

mass of a proton = 1.00728 u

mass of a neutron = 1.00867 u

mass of ${}^7_3\text{Li}$ nucleus = 7.01436 u

[1 mark]

A 0.93912 u

☐

B 0.04051 u

☐

C 0.04077 u

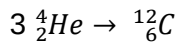
☐

D 0.04216 u

☐

2	2
---	---

One of the possible steps in the fusion of Helium in the sun is shown below:



The mass of a Helium -4 nucleus is 4.001505 u

The mass of a Carbon-12 nucleus is 12.000000 u

Calculate the energy released in this process in Joules.

[1 mark]

A 6.73×10^{-13} J

☐

B 3.97×10^{-10} J

☐

C 1.19×10^{-9} J

☐

D 2.45×10^{-9} J

☐

2	3
---	---

A company makes small solar cells for use in toys. They make two types. Both are square and the sides of Type A are 4 times as long as those for Type B. The efficiency of both cells is the same. In a test, a Type A cell is positioned 20 cm from a light source.

At what distance from the source will the Type B cell have the same output power?
[1 mark]

A 5 cm



B 10 cm



C 100 cm



D 160 cm

